

Expt 10	Digital Speech Processing and Noise Reduction
Date:	10.1 Implementation of Spectral Subtraction for Speech Noise Reduction

Code:

```

clc;
clear;
close all;

%% Parameters
Fs = 8000;           % Sampling frequency
t = 0:1/Fs:3;        % Time vector

%% Clean Speech Signal (Example)
x = sin(2*pi*200*t) + 0.5*sin(2*pi*400*t);

%% Add White Gaussian Noise
noise = 0.3*randn(size(x));
xn = x + noise;

%% FFT of Noisy Signal and Noise
Y = fft(xn);
N = fft(noise);

%% Spectral Subtraction
MagY = abs(Y);
MagN = abs(N);
PhaseY = angle(Y);

MagE = max(MagY - MagN,0);

Xe = MagE .* exp(1j*PhaseY);
xe = real(ifft(Xe));

%% SNR Calculation
SNR_before = 10*log10(sum(x.^2)/sum((xn-x).^2));
SNR_after = 10*log10(sum(x.^2)/sum((xe-x).^2));

fprintf('Input SNR = %.2f dB\n',SNR_before);
fprintf('Output SNR = %.2f dB\n',SNR_after);

%% Time Domain Signals
figure;

subplot(3,1,1)
plot(t,x)
grid on
title('Clean Signal')
xlabel('Time (s)')

```

```
ylabel('Amplitude')
```

```
subplot(3,1,2)  
plot(t,xn)  
grid on  
title('Noisy Signal')  
xlabel('Time (s)')  
ylabel('Amplitude')
```

```
subplot(3,1,3)  
plot(t,xe)  
grid on  
title('Enhanced Signal')  
xlabel('Time (s)')  
ylabel('Amplitude')
```

```
%% Spectrograms
```

```
figure;
```

```
subplot(3,1,1)  
spectrogram(x,256,200,256,Fs,'yaxis')  
title('Clean Spectrogram')
```

```
subplot(3,1,2)  
spectrogram(xn,256,200,256,Fs,'yaxis')  
title('Noisy Spectrogram')
```

```
subplot(3,1,3)  
spectrogram(xe,256,200,256,Fs,'yaxis')  
title('Enhanced Spectrogram')
```

```
%% Magnitude Spectrum
```

```
f = (0:length(x)-1)*Fs/length(x);
```

```
figure;
```

```
subplot(3,1,1)  
plot(f,abs(fft(x)))  
xlim([0 Fs/2])  
grid on  
title('Clean Spectrum')  
xlabel('Frequency (Hz)')  
ylabel('Magnitude')
```

```
subplot(3,1,2)  
plot(f,abs(fft(xn)))  
xlim([0 Fs/2])  
grid on  
title('Noisy Spectrum')  
xlabel('Frequency (Hz)')
```

```
ylabel('Magnitude')
```

```
subplot(3,1,3)  
plot(f,abs(fft(xe)))  
xlim([0 Fs/2])  
grid on  
title('Enhanced Spectrum')  
xlabel('Frequency (Hz)')  
ylabel('Magnitude')
```

Output:

Fig 1:

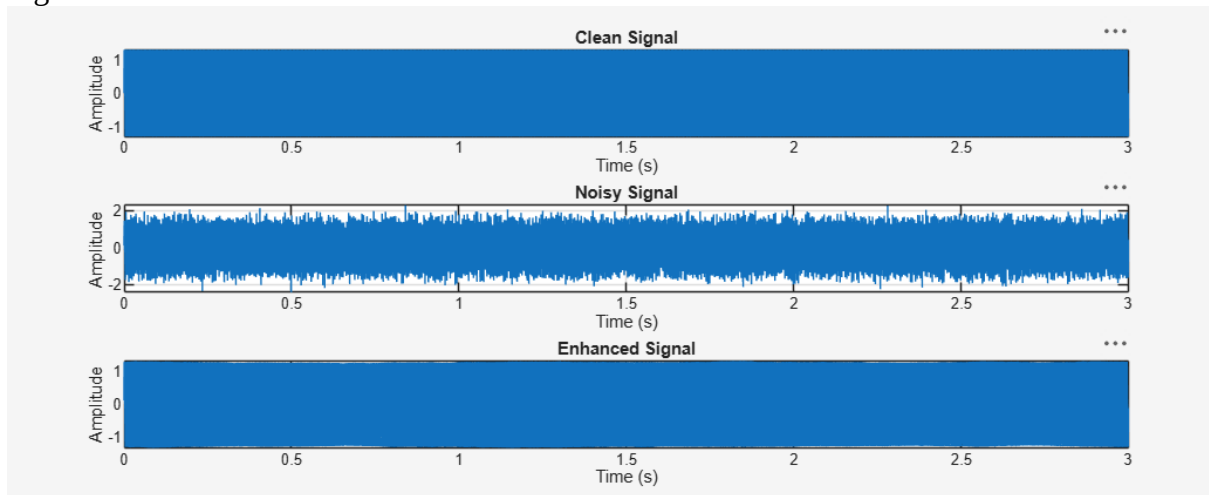


Fig 2:

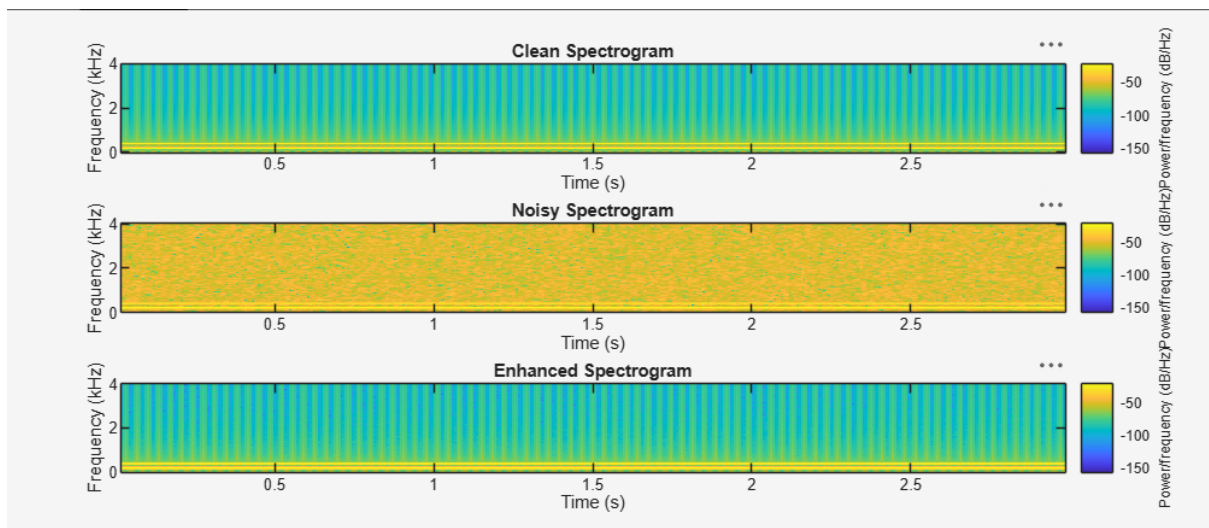


Fig 3:

